IAT New Test 9

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

Instructions

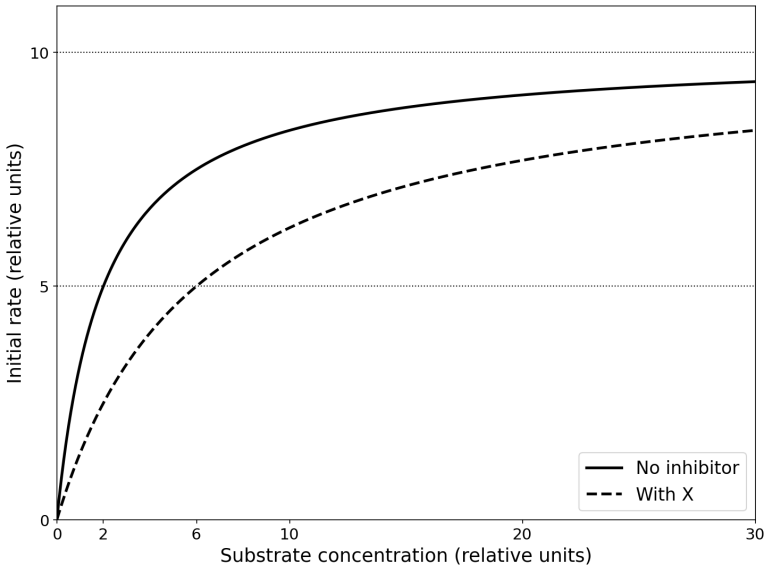
Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

Q1 A fern bears one type of spore, and a free-living prothallus develops from it. Which conclusion follows?
A) It is homosporous; its prothallus is the gametophyte
B) It is heterosporous; the prothallus is sporophyte
C) Its spores are diploid gametes
D) Its prothallus makes spores by meiosis

Q2 A floral formula records K(5) C(5) A5 G̲(2), with the underline denoting a superior ovary. Which interpretation is consistent?
A) Free sepals and two free carpels
B) Five fused sepals, five fused petals and a bicarpellary syncarpous superior ovary
C) An inferior ovary and ten stamens
D) A monocarpellary ovary with free petals

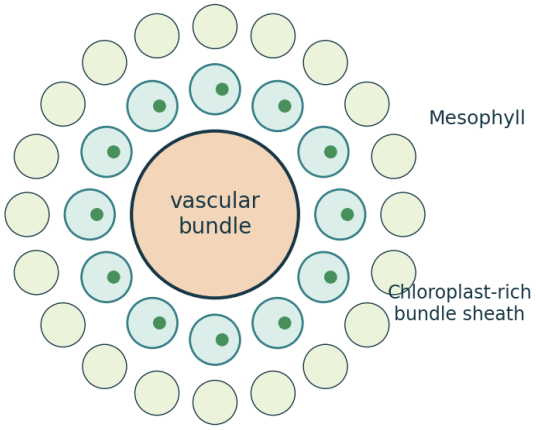
Q3 The enzyme-rate figure compares an uninhibited curve with the curve under inhibitor X. Which inhibition mechanism best fits both Km and Vmax?



A) Irreversible loss of enzyme
B) Uncompetitive inhibition lowering both parameters
C) Competitive inhibition that can be overcome by high substrate
D) Activation of catalytic turnover

Q4 In the leaf cross-section, bundle-sheath cells form a chloroplast-rich ring around each vascular bundle and mesophyll surrounds the ring. Which conclusion best fits?

Leaf bundle and surrounding cells



A) CAM storage is certain
B) Only C3 fixation can occur
C) The bundle sheath lacks chloroplasts
D) Kranz anatomy supports spatial C4 carbon concentration

Q5 A postsynaptic neuron responds to externally applied transmitter but not to a presynaptic action potential. Its axon still conducts when directly stimulated. What is the most localised defect?
A) Transmitter release at the presynaptic terminal
B) Loss of all postsynaptic receptors
C) Failure of axonal Na^+ channels throughout the postsynaptic neuron
D) Inability of the transmitter to bind the intact receptor

Q6 After fertilisation in an angiosperm, what do an ovule and an ovary generally become, respectively?
A) Embryo and endosperm
B) Seed and fruit
C) Fruit and seed
D) Seed coat and pollen grain

Q7 An X-linked recessive disorder affects a father and not his partner, who is homozygous normal. What is expected among their children?
A) Every son affected
B) Half of sons affected
C) Every daughter a carrier and every son unaffected
D) Every daughter affected

Q8 A cell culture loses p53 function. Following DNA damage it continues through G1/S and accumulates mutations. What best explains its greater cancer risk?
A) It automatically gains a new chromosome each division
B) Repair becomes unnecessary
C) Mitosis becomes meiosis
D) The damage checkpoint fails to pause or eliminate compromised cells

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Q9 A plasmid has ampicillin resistance and a disrupted tetracycline-resistance gene when insert is present. Transformants first grow on ampicillin. What screen selects likely recombinants?
A) Replica plate and retain colonies that fail to grow on tetracycline
B) Keep only colonies growing on both drugs
C) Keep cells that never grew on ampicillin
D) Discard all ampicillin-resistant colonies

Q10 In an energy-flow measurement, producers fix 10,000 kJ; 2,000 kJ enters herbivores, which lose 1,200 kJ as heat and pass 300 kJ to predators. How much herbivore energy enters detritus?
A) 300 kJ
B) 500 kJ
C) 800 kJ
D) 1,500 kJ

Q11 A replication fork advances right. The upper template runs 3′→5′ toward the fork, and its new daughter extends continuously toward the fork. Which statement is correct?
A) New DNA grows 3′→5′
B) Both daughters extend continuously in the fork direction
C) The upper daughter is leading and grows 5′→3′
D) The lower daughter must be synthesized 3′→5′

Q12 In the lac-operon table, strain X expresses lacZ without lactose but shuts off in glucose; strain Y responds normally to both conditions. X has a repressor that cannot bind operator but intact glucose control. What explains X?
A) A defective promoter preventing all transcription
B) An enzyme unable to cleave lactose
C) A permanently inactive polymerase
D) Constitutive operator-level derepression with catabolite repression retained

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Q13 A time-series measurement shows that a rise in circulating thyroid hormone is followed by a fall in TSH while external conditions are unchanged. Which feedback link is supported?
A) Thyroid hormone negatively feeds back on pituitary TSH output
B) TSH suppresses thyroid hormone irreversibly
C) Thyroid hormone is made by the pituitary
D) The thyroid axis has only positive feedback

Q14 A ribosome can form a peptide bond with already charged tRNAs, but an inhibitor prevents the mRNA–tRNA complex from advancing to expose the next codon. Which stage is most directly blocked?
A) Transcription initiation
B) Translocation during translation elongation
C) Amino-acid activation
D) Termination before any peptide bond

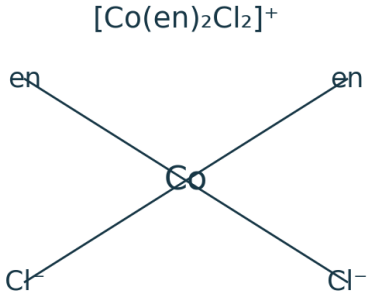
Q15 For the same habitat type, species richness follows $S = cA^{0.25}$. If sampled area rises by a factor of 16 with c unchanged, by what factor does expected richness rise?
A) 4
B) 8
C) 16
D) 2

Chemistry

Q16 The H–O–H angle is about 104.5°, below the tetrahedral ideal. Which electron-pair explanation is best?
A) No lone pair on oxygen
B) Three bonding pairs on oxygen
C) Two lone pairs exert stronger repulsion than bonding pairs and compress the angle
D) The molecule is linear but rotates rapidly

Q17 Use the coordination sketch and its donor-site labels to find the metal oxidation state and coordination number.

Coordination sphere and donor sites



Each en makes two donor bonds; each Cl^- makes one

A) +2 and 4
B) +2 and 6
C) +3 and 4
D) +3 and 6

Q18 Which ion is d^0 and hence lacks d–d transitions within its own d subshell?
A) Sc^{3+}
B) Ti^{3+}
C) V^{3+}
D) Cr^{3+}

Q19 A hydrogen atom emits a photon in the 3→2 transition. If $E_n = -13.6/n^2$ eV and $hc = 1240$ eV·nm, its wavelength is closest to?
A) 486 nm
B) 656 nm
C) 434 nm
D) 122 nm

Q20 Why can the first ionisation energy of oxygen be lower than that of nitrogen despite oxygen being to its right in period 2?
A) Oxygen has a completely empty 2p shell
B) Nitrogen has a paired 2p orbital
C) Oxygen has a paired 2p electron; nitrogen has a more stable half-filled $2p^3$ arrangement
D) Nitrogen has a larger principal quantum number

Q21 A 10.0 g sample contains $CaCO_3$ and an inert impurity that does not react with acid. Excess HCl releases 1.76 g CO_2, all collected. With $M(CaCO_3) = 100$ and $M(CO_2) = 44$ g mol^{-1}, what is the sample's $CaCO_3$ mass fraction?
A) 0.40
B) 0.176
C) 0.60
D) 0.80

Q22 Compare p-nitrobenzoic acid and p-methoxybenzoic acid in water. Which is generally more acidic, and why?
A) Methoxy acid: +M donation stabilises its anion
B) Nitro acid: electron withdrawal stabilises its conjugate base
C) Equal: para groups cannot affect resonance
D) Methoxy acid: inductive donation alone dominates

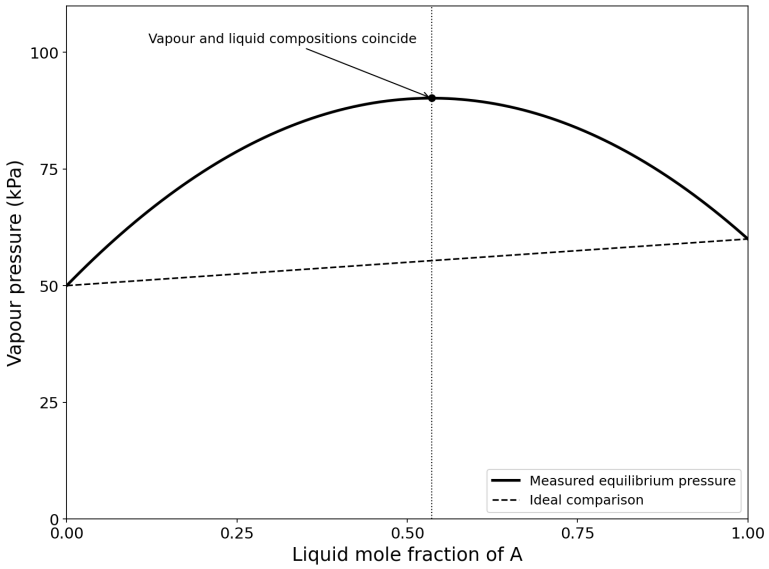
Q23 Which single compound can undergo self-aldol addition under basic conditions?
A) Benzaldehyde
B) Formaldehyde
C) Propanal
D) Benzophenone

Q24 An organic solution contains benzoic acid, phenol and anisole. It is extracted first with aqueous $NaHCO_3$, then its remaining organic layer is extracted with aqueous NaOH. Assume sufficient reagent, efficient separation and negligible neutral-solute transfer. On acidifying only the second aqueous extract, which organic compound is recovered?
A) Benzoic acid only
B) Anisole only
C) Benzoic acid and phenol
D) Phenol only

Q25 An unknown amine reacts with benzenesulfonyl chloride in alkali to give a sulfonamide insoluble in alkali; it does not dissolve after acidification. Which class is most consistent?
A) Secondary amine
B) Primary amine
C) Tertiary amine
D) Ammonium salt without an amine

Q26 At a pH far above its isoelectric point, a simple amino acid without an ionisable side chain predominantly has what net charge?
A) +2
B) −1
C) 0
D) +1

Q27 The vapour-pressure/composition diagram at fixed temperature shows an interior maximum where vapour and liquid compositions coincide. Which boiling and simple-distillation behaviour is consistent with this azeotrope?



A) Maximum boiling temperature; distillation always separates it into pure components
B) Minimum boiling temperature; distillation changes its composition continuously
C) Minimum boiling temperature; it distils at unchanged composition
D) Maximum boiling temperature; no vapour forms at that composition

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Q28 A concentration cell $Cu|Cu^{2+}(0.010\ M)||Cu^{2+}(1.0\ M)|Cu$ operates at 298 K. With E° identical and 0.0591 V in the Nernst factor, what is Ecell?
A) 0.0296 V
B) 0.0591 V
C) 0.1182 V
D) 0 V

Q29 For a reaction with Ea=50 kJ mol^{-1}, what is approximately k(310 K)/k(300 K)? Take R=8.314 J mol^{-1} K^{-1}.
A) 1.91
B) 1.09
C) 1.48
D) 4.8

Q30 For $N_2+3H_2\rightleftharpoons 2NH_3$ at fixed T, the initial quotient Q is less than K. What is the immediate spontaneous direction?
A) Reverse, because ammonia is a product
B) No change, because T is fixed
C) Both directions equally
D) Forward until Q approaches K

Mathematics

Q31 Let A be invertible 3×3 with $\det(A)=3$. Form B by adding twice row 1 to row 2, then multiplying row 3 by -2. What is $\det(\text{adj } B)$?
A) -36
B) -6
C) 36
D) 6

Q32 On a square grid, how many right/up paths go from (0,0) to (3,3) without passing through (1,1)?
A) 12
B) 10
C) 6
D) 8

Q33 For real x, which inputs make $\ln(x^2-2)$ defined if this expression is obtained by $f(g(x))$, with $f(t)=\ln(t-1)$ and $g(x)=x^2-1$?
A) $(-\infty,-\sqrt{2})\cup(\sqrt{2},\infty)$
B) $[-\sqrt{2},\sqrt{2}]$
C) All nonzero x
D) $(-\infty,-1)\cup(1,\infty)$

Q34 In the Argand plane, what locus satisfies $|z-(1+i)|=|z-(3-i)|$?
A) $x+y=2$
B) $x-y=2$
C) $x+y=1$
D) $x-y=1$

Q35 Evaluate $\int_0^{\pi} x \sin x\, dx$.
A) $\pi/2$
B) 2π
C) π
D) 0

Q36 Project the point P=(1,2) orthogonally onto the oblique line $3x+4y-10=0$ in the plane. Which ordered pair is obtained?
A) (4/5,9/5)
B) (0,5/2)
C) (1,7/4)
D) (22/25,46/25)

Q37 For $f(x)=ax+b$ if $x<1$, and x^2 if $x\geq1$, which (a,b) make f differentiable at $x=1$?
A) $(2,-1)$
B) (1,0)
C) (2,1)
D) $(-2,3)$

Q38 For the observations 1, 1 and 4, what is the mean absolute deviation about their median?
A) 4/3
B) 1
C) 2
D) 3

Q39 Find a such that $\lim(x\to0)\ (\sin(ax)-2x)/x^3$ exists as a finite real number.
A) 0
B) 1
C) 2
D) 3

Q40 Urn U_1 is chosen with probability 1/3 and has 2 red, 1 blue; U_2 with probability 2/3 has 1 red, 3 blue. After observing red, what is $P(U_1|\text{red})$?
A) 1/3
B) 2/3
C) 1/2
D) 4/7

Q41 If $\tan\theta=1/2$ with θ acute, find $\tan(2\theta)$.
A) 4/3
B) 3/4
C) 1
D) 2

Q42 For $x>0$ and $y>0$, solve $dy/dx=(x^2+y^2)/(xy)$ with $y(1)=1$. What is $y(e)$?
A) $e\sqrt{2}$
B) $e\sqrt{3}$
C) $2e$
D) $3e$

Q43 For which real k are both roots of $x^2-2kx+(k+3)=0$ real and in the open interval (0,3)?
A) $k\geq(1+\sqrt{13})/2$
B) $(1+\sqrt{13})/2<k\leq12/5$
C) $(1+\sqrt{13})/2\leq k<12/5$
D) $k>12/5$

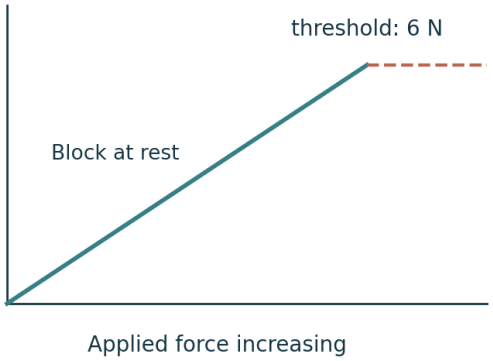
Q44 A line meets the positive coordinate axes at $A(a,0)$ and $B(0,b)$, and passes through (2,3). What is the least possible area of triangle OAB, where O is the origin?
A) 6
B) 8
C) 10
D) 12

Q45 For which k are (1,2), (3,6), and (k,10) collinear?
A) 4
B) 5
C) 6
D) 7

Physics

Q46 The figure gives the static-friction threshold for a block on a horizontal floor. While the block remains at rest under a horizontal 4 N applied force, what is its friction?

Applied force and static-friction threshold



A) 4 N opposite the applied force
B) 6 N opposite the force
C) 2 N opposite the force
D) Zero until sliding

Q47 A one-dimensional conservative force is $F(x)=x^2-4x$ newtons, with x measured in metres. What is its signed work as a particle moves from $x=0$ to $x=4$ m?
A) $+32/3$ J
B) $-32/3$ J
C) 0
D) -16 J

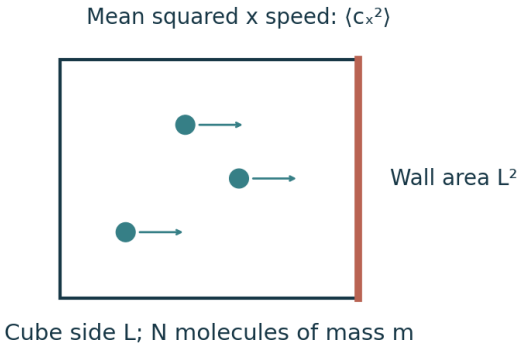
Q48 At a planet surface $r=R$, escape speed is v. What minimum launch speed reaches $r=4R$ with zero speed there, ignoring atmosphere and rotation?
A) $v/2$
B) $v/4$
C) $\sqrt{3}v/2$
D) $3v/4$

Q49 A stretched rod has longitudinal strain $+0.002$ and Poisson ratio 0.30. What is the transverse strain?
A) $+0.006$
B) -0.006
C) $+0.0006$
D) -0.0006

Q50 Two identical hot bodies at 80°C and 60°C cool in air at 20°C under the same Newton cooling constant. What is the ratio of their initial cooling-rate magnitudes, hotter to cooler?
A) 3:2
B) 2:1
C) 4:3
D) 1:1

Q51 The kinetic-theory sketch shows N molecules of mass m in a cube of side L, each with mean-square x-velocity component $\langle c_x^2 \rangle$. What pressure acts on the wall normal to x?

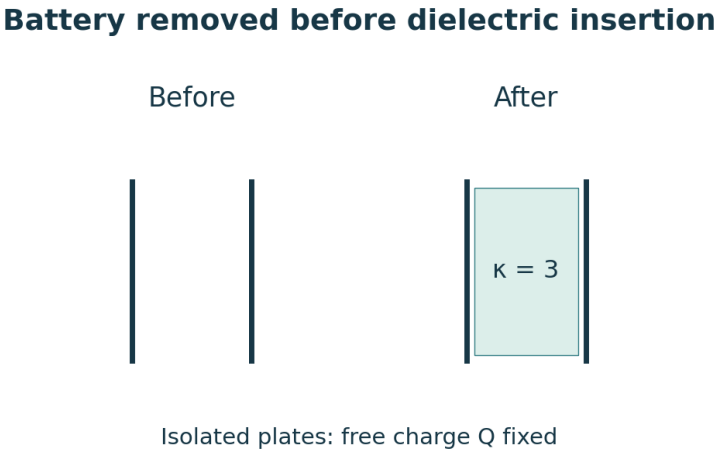
Molecular impacts on the x-normal wall



A) $Nm\langle c_x^2 \rangle/L^2$
B) $Nm\langle c_x^2 \rangle/L^3$
C) $Nm\langle c_x^2 \rangle/(3L^3)$
D) $Nm\langle c_x^2 \rangle/(6L^3)$

Q52 For SHM with total mechanical energy E and amplitude A, what is kinetic energy at displacement x=A/2?
A) E/4
B) E/2
C) 3E/4
D) E

Q53 The before/after diagram shows a charged capacitor isolated from the battery and then filled with the indicated dielectric. What are its new voltage and energy relative to V_0,U_0?



A) $3V_0$,$3U_0$
B) V_0,U_0
C) $V_0/3$,$3U_0$
D) $V_0/3$,$U_0/3$

Q54 A positive charge moves with velocity v $\hat{x}$ through B($\hat{y}$+$\hat{z}$). Which uniform electric field makes its total Lorentz force zero?
A) vB($\hat{y}-\hat{z}$)
B) vB($\hat{z}-\hat{y}$)
C) vB($\hat{y}+\hat{z}$)
D) Zero

Q55 An isolated conducting shell has a cavity containing a point charge +q and carries total charge −2q. An additional charge −5q is outside the shell. In electrostatic equilibrium, what are the total charges on the cavity surface and outer surface, and the flux through a closed surface enclosing the shell and cavity but not the external charge?
A) $-q;\ -2q;\ -2q/\varepsilon_0$
B) $-q;\ -q;\ -q/\varepsilon_0$
C) $0;\ -2q;\ -q/\varepsilon_0$
D) $+q;\ -3q;\ -6q/\varepsilon_0$

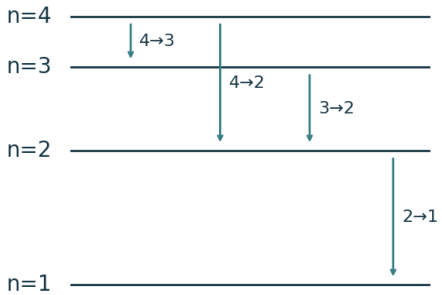
Q56 A resistor network has two identical 6 Ω resistors in parallel between A and B, then a 2 Ω resistor in series from B to C. What is equivalent resistance between A and C?
A) 3 Ω
B) 4 Ω
C) 5 Ω
D) 8 Ω

Q57 An object 30 cm left of a converging lens (f=10 cm) first forms an image; a plane mirror 20 cm right of the lens reflects those rays back through the lens. Where is the final image after the second lens passage?
A) 5 cm left of lens
B) 10 cm left of lens
C) 15 cm left of lens
D) 50/3 cm left of lens

Q58 At fixed frequency above threshold, light intensity incident on a photocathode doubles while collector voltage is high enough to saturate. What happens to saturation photocurrent and stopping potential?
A) Current doubles; stopping potential unchanged
B) Both double
C) Current unchanged; potential doubles
D) Both unchanged

Q59 The hydrogen-level figure marks n=4,3,2,1. Which downward transition has the longest emitted wavelength among 4→3, 4→2, 3→2, 2→1?

Hydrogen energy levels



A) 4→2
B) 3→2
C) 4→3
D) 2→1

Q60 Two coils have positive mutual inductance M, with signed flux linkage in the secondary defined as $\Lambda_2 = MI_1$. Its closed resistance is R; neglect secondary self-inductance. The primary current is $I_1(t) = I_0(1 - t/T)^2$ for $0 \le t \le T$. With positive secondary emf defined by $\mathcal{E}_2 = -d\Lambda_2/dt$, what signed charge passes in the secondary during $0 \le t \le T/2$?
A) $-3MI_0/(4R)$
B) $MI_0/(4R)$
C) $MI_0/(2R)$
D) $3MI_0/(4R)$

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / A pteridophyte spore type

Answer: (A) It is homosporous; its prothallus is the gametophyte.

Why: A single spore class defines homospory. Germination gives the haploid prothallus bearing gametangia; the spore-bearing leafy fern is the sporophyte. This identifies the free-living stage from its origin and reproductive structures rather than from its size.

Why not the others: (B) invents two spore classes and mislabels the generation; (C) spores are products of meiosis, not gametes; (D) spore meiosis belongs to the sporophyte sporangium.

Key point: Track ploidy and generation separately.

Q2 / B floral formula inference

Answer: (B) Five fused sepals, five fused petals and a bicarpellary syncarpous superior ovary.

Why: Parentheses indicate fusion within a whorl: K(5), C(5), and G(2) are fused groups. A5 means five stamens; the stated convention marks the ovary superior.

Why not the others: (A) ignores the parentheses; (C) changes both ovary position and stamen count; (D) contradicts petal and carpel fusion.

Key point: Read the symbols and position convention independently.

Q3 / C enzyme inhibition

Answer: (C) Competitive inhibition that can be overcome by high substrate.

Why: The same asymptotic Vmax means sufficiently abundant substrate restores the maximal rate, while a larger substrate concentration is needed for half-maximal rate. Competition for the active site has this pattern.

Why not the others: (A) usually lowers functional enzyme and Vmax; (B) changes both Km and Vmax downward; (D) would increase maximal output rather than preserve it.

Key point: Compare both Km and Vmax, not a single rate.

Q4 / D C3 versus C4 anatomy

Answer: (D) Kranz anatomy supports spatial C4 carbon concentration.

Why: The pictured concentric organisation is Kranz anatomy. Initial fixation in mesophyll can be followed by CO_2 release near RuBisCO in bundle sheath, reducing oxygenation under suitable conditions.

Why not the others: (A) CAM relies on temporal rather than this spatial separation; (B) conflicts with the usual C4 interpretation; (C) contradicts the drawn chloroplasts.

Key point: Anatomy supports a pathway inference, not a measured rate.

Q5 / A neural synapse and receptor

Answer: (A) Transmitter release at the presynaptic terminal.

Why: Direct transmitter application verifies that postsynaptic receptor and downstream excitability can work. Direct axon stimulation verifies conduction. The missing response only when presynaptic input is used points to release or delivery from that terminal.

Why not the others: (B) and (D) conflict with the direct application response; (C) conflicts with direct axonal conduction.

Key point: Use bypass experiments to locate a failed step.

Q6 / B seed and fruit development

Answer: (B) Seed and fruit.

Why: The fertilised ovule matures as a seed containing the embryo; the enclosing ovary generally develops into the fruit. Integuments contribute the seed coat, but that is only one part of the seed.

Why not the others: (A) names contents rather than the enclosing structures; (C) reverses the two; (D) substitutes a substructure and an unrelated male gametophyte.

Key point: Follow the whole structures through maturation.

Q7 / C sex-linked transmission

Answer: (C) Every daughter a carrier and every son unaffected.

Why: The father gives his Y to sons and affected X to daughters. The mother supplies a normal X to each child; daughters are heterozygous carriers and sons receive their only X from her.

Why not the others: (A) and (B) incorrectly send the paternal X to sons; (D) overlooks the normal maternal X.

Key point: Separate paternal transmission by child sex.

Q8 / D cancer and cell-cycle link

Answer: (D) The damage checkpoint fails to pause or eliminate compromised cells.

Why: p53 supports arrest and downstream repair or apoptosis responses after damage. Without that restraint, damaged genomes enter replication and pass variants to descendants, increasing opportunity for malignant selection.

Why not the others: (A) is not automatic; (B) repair remains needed; (C) changes the division type without support.

Key point: A failed checkpoint permits propagation of damage.

Q9 / A recombinant selection control

Answer: (A) Replica plate and retain colonies that fail to grow on tetracycline.

Why: Ampicillin growth selects plasmid-containing cells. A correctly inserted fragment disables tetracycline resistance, so ampicillin-positive but tetracycline-negative colonies are candidate recombinants; verify insert by a separate assay.

Why not the others: (B) favours intact vector; (C) lacks evidence of plasmid uptake; (D) discards the selected transformants.

Key point: Selection and insertional screening answer different questions.

Q10 / B food-web energy accounting

Answer: (B) 500 kJ.

Given: Input 2000 kJ and two known outputs.

Formula/Principle: Energy balance: detritus = input − heat − predation.

Step 1: 2000−1200−300=500 kJ.

Why: Conservation across the herbivore box gives 2,000 kJ in minus 1,200 kJ heat minus 300 kJ predator transfer, leaving 500 kJ for detritus. This bookkeeping does not assume an arbitrary 10% rule.

Why not the others: (A) repeats predator transfer; (C) subtracts only heat; (D) subtracts only detrital or trophic routes incorrectly.

Key point: Account for every output of the same compartment.

Q11 / C molecular replication direction

Answer: (C) The upper daughter is leading and grows 5′→3′.

Why: DNA polymerase extends the 3′ end, giving synthesis 5′→3′. Reading the upper template 3′→5′ toward the advancing fork allows continuous leading-strand synthesis in that direction; the other template requires fragments.

Why not the others: (A) reverses polymerase direction; (B) ignores antiparallel templates; (D) reverses synthesis direction on the lagging strand.

Key point: The 5′→3′ synthesis rule applies to both daughters.

Q12 / D transcription regulation

Answer: (D) Constitutive operator-level derepression with catabolite repression retained.

Why: A repressor that cannot occupy operator cannot shut the operon off in the absence of inducer. The intact glucose control still lowers activation under glucose, explaining both observed features.

Why not the others: (A) and (C) predict no expression; (B) concerns substrate processing rather than operator occupancy and does not explain constitutive transcription.

Key point: Two regulatory layers can act independently.

Q13 / A endocrine feedback

Answer: (A) Thyroid hormone negatively feeds back on pituitary TSH output.

Why: An increase in the downstream hormone followed by a reduced upstream drive under otherwise stable conditions is the expected sign of negative feedback. The graph alone does not identify every molecular intermediate.

Why not the others: (B) reverses the observed causal direction; (C) misplaces the hormone source; (D) predicts TSH increasing rather than falling.

Key point: Use direction and sequence of change to identify feedback sign.

Q14 / B Localising the translocation stage of translation

Answer: (B) Translocation during elongation.

Why: The charged tRNAs and successful peptide-bond formation place the block after amino-acid activation and the bond-forming part of elongation. Moving the mRNA–tRNA complex relative to the ribosome is translocation; without it, the next codon cannot be presented for another elongation cycle. Transcription is a separate DNA-to-RNA process and does not move a completed mRNA through the ribosome.

Why not the others: (A) concerns making RNA, not advancing it during translation; (C) has already supplied the charged tRNAs; (D) describes ending the chain rather than the inhibited codon-to-codon movement.

Key point: Use events that still occur to locate the first blocked step.

Q15 / D Species-area scaling with a common habitat coefficient

Answer: (D) 2.

Given: $A_2/A_1 = 16$ and the fitted exponent is $1/4$.

Formula/Principle: Take the ratio so the common factor c cancels.

Step 1:

$$\frac{S_2}{S_1} = \left(\frac{A_2}{A_1}\right)^{1/4}.$$

Step 2:

$$\frac{S_2}{S_1} = 16^{1/4} = 2.$$

Why: The model is sublinear: increasing area sixteenfold does not increase the expected species count sixteenfold. The same habitat coefficient must apply to both areas for it to cancel in this comparison.

Why not the others: (A) uses exponent one half; (B) uses exponent three quarters; (C) assumes direct proportionality with exponent one.

Key point: Match each proposed scale factor to the exponent it actually uses.

Chemistry

Q16 / C VSEPR angle with lone pair

Answer: (C) Two lone pairs exert stronger repulsion than bonding pairs and compress the angle.

Why: Oxygen has two O–H bond pairs and two lone pairs in approximately tetrahedral electron-pair geometry. Lone-pair repulsions push bonding pairs closer, reducing the observed bond angle.

Why not the others: (A) omits two lone pairs; (B) counts bonds incorrectly; (D) contradicts bent molecular geometry.

Key point: Distinguish electron-pair geometry from molecular shape.

Q17 / D coordination number and charge

Answer: (D) +3 and 6.

Why: Let Co oxidation state be x. Two neutral en and two Cl^- give x−2=+1, so x=+3. Each en donates two sites (four total) and two chlorides one each, giving coordination number six.

Why not the others: (A) and (B) miscount charge; (C) counts ligand molecules rather than donor atoms.

Key point: Charge and donor count require separate calculations.

Q18 / A colour and d-electron count

Answer: (A) Sc^{3+}.

Why: Neutral Sc is $[Ar]3d^14s^2$; removal of three electrons gives $[Ar]3d^0$. With no d electron, a d–d excitation is unavailable, though other colour mechanisms can exist in compounds.

Why not the others: (B) Ti^{3+} is d^1; (C) V^{3+} is d^2; (D) Cr^{3+} is d^3.

Key point: Compute the ionic d count before invoking colour.

Q19 / B hydrogen transition wavelength

Answer: (B) 656 nm.

Given: Hydrogen levels n=3 and 2.

Formula/Principle: ΔE=13.6(1/2²−1/3²) eV; λ=hc/ΔE.

Step 1: ΔE=1.889 eV.

Worked derivation:

$$\Delta E = 13.6(1/4 - 1/9) = 1.889 \text{ eV},$$
$$\lambda = hc/\Delta E = 1240/1.889 \approx 656.5 \text{ nm}.$$

Why: The energy gap is 13.6(1/4−1/9)=1.889 eV, so λ=1240/1.889≈656 nm. The emitted photon energy is the positive difference of level energies. Since wavelength varies inversely with energy, choosing a larger transition gap would give a shorter line.

Why not the others: (A) and (C) correspond to different Balmer transitions; (D) is much shorter wavelength and requires a larger gap.

Key point: Use energy difference, then convert by hc/E.

Q20 / C ionisation anomaly

Answer: (C) Oxygen has a paired 2p electron; nitrogen has a more stable half-filled 2p³ arrangement.

Why: Nitrogen 2p³ puts one electron in each p orbital; oxygen 2p⁴ introduces a pair. Repulsion in that pair eases removal relative to the half-filled nitrogen configuration, overcoming the simple across-period trend locally.

Why not the others: (A) reverses configuration; (B) gives the pair to nitrogen; (D) both occupy n=2.

Key point: Local subshell stability modifies periodic trends.

Q21 / A Composition from selective carbonate gas evolution

Answer: (A) 0.40.

Given: Only $CaCO_3$ contributes to the collected gas and HCl is in excess.

Formula/Principle:

$$CaCO_3 + 2\,HCl \longrightarrow CaCl_2 + CO_2 + H_2O.$$

Step 1:

$$n_{CO_2} = 1.76/44 = 0.0400 \text{ mol}.$$

Step 2: The 1:1 carbon balance gives $n_{CaCO_3} = 0.0400$ mol and mass $= 0.0400(100) = 4.00$ g.

Step 3:

$$w_{CaCO_3} = \frac{4.00}{10.0} = 0.400.$$

Why: Collected carbon dioxide measures the reacting carbonate rather than the whole sample mass. Excess acid and an inert impurity make carbonate the sole limiting source of gas, so an isolation-yield correction is not appropriate.

Why not the others: (B) divides gas mass directly by sample mass; (C) reports the impurity fraction; (D) inserts a false 2:1 carbonate-to-gas ratio.

Key point: Convert a selective reaction signal to the original component's mass.

Q22 / B resonance versus inductive acidity

Answer: (B) Nitro acid: electron withdrawal stabilises its conjugate base.

Why: The para nitro group withdraws electron density through resonance and induction, stabilising the negatively charged benzoate relative to the acid. Para methoxy donates through resonance and tends to oppose this stabilisation.

Why not the others: (A) calls electron donation stabilising for an anion; (C) ignores conjugation through the ring; (D) reverses the key para resonance effect.

Key point: Judge acidity by relative conjugate-base stability.

Q23 / C aldol eligibility

Answer: (C) Propanal.

Why: Propanal has an alpha carbon bearing hydrogens, so enolate formation and attack on another propanal molecule are possible. The listed aromatic or one-carbon carbonyl compounds lack alpha hydrogen in the needed position.

Why not the others: (A) benzaldehyde lacks an alpha carbon; (B) formaldehyde lacks one; (D) benzophenone has phenyl groups directly attached to carbonyl carbon, without alpha hydrogen.

Key point: Check for at least one alpha hydrogen on the enolate donor.

Q24 / D Sequential acid-base extraction of phenol

Answer: (D) Phenol only.

Why: Bicarbonate first converts the stronger benzoic acid to a water-soluble carboxylate, removing it before the next extraction. Phenol is too weak an acid for that bicarbonate wash but is deprotonated by NaOH and enters the second aqueous layer as phenoxide. Acidifying that separated layer regenerates phenol; anisole has no acidic O–H proton and remains neutral in the stated washes.

Step 1: First wash removes benzoic acid as benzoate.

Step 2: Second wash removes phenol as phenoxide.

Step 3: Acidification of the second wash restores phenol.

Why not the others: (A) belongs to the first extract; (B) assumes anisole has an acidic O–H group and forms a water-soluble phenoxide; (C) ignores separation of the first aqueous wash.

Key point: Track acid strength and layer history through successive extractions.

Q25 / A amine-class test

Answer: (A) Secondary amine.

Why: A secondary amine forms a sulfonamide with no N–H remaining, so it does not form the soluble anion in alkali. A primary sulfonamide retains an acidic N–H and dissolves in alkali; a tertiary amine lacks the N–H needed to form this sulfonamide.

Why not the others: (B) product should dissolve in base; (C) cannot furnish the described sulfonamide; (D) does not explain the reaction product.

Key point: Hinsberg solubility depends on residual sulfonamide N–H.

Q26 / B amino acid charge

Answer: (B) -1.

Why: High pH deprotonates the carboxyl to COO^- and the ammonium group to neutral NH_2, giving a net -1. This describes the predominant form sufficiently above both relevant pKa values.

Why not the others: (A) overcounts positive charge; (C) is the zwitterion near its isoelectric region; (D) is favoured at strongly acidic pH.

Key point: Compare pH with protonation equilibria of both groups.

Q27 / C Pressure maximum and azeotropic distillation

Answer: (C) A minimum-boiling azeotrope that distils without a composition change.

Why: A composition with unusually high vapour pressure reaches a given external pressure at a lower temperature, giving the minimum-boiling tendency. At the azeotrope the vapour has the same composition as the liquid, so removing that vapour does not enrich either component by ordinary simple distillation. This conclusion uses the interior maximum and the equality of phase compositions, not merely the sign of a deviation at one arbitrary composition.

Why not the others: (A) reverses the boiling tendency and ignores azeotropic phase equality; (B) contradicts equal vapour/liquid composition; (D) reverses the pressure–boiling relation and incorrectly denies vaporisation.

Key point: Connect a pressure extremum to boiling tendency and equal phase composition.

Q28 / B concentration-cell potential

Answer: (B) 0.0591 V.

Given: Cu^{2+} concentrations differ by factor 100.

Formula/Principle: E=(0.0591/n) log(cathode/anode), n=2.

Step 1: E=0.02955×2=0.0591 V.

Worked derivation:

$$E = \frac{0.0591}{2} \log_{10} \frac{1.0}{0.010} = 0.02955(2) = 0.0591 \text{ V}.$$

Why: The higher-concentration copper(II) half-cell is reduced and the lower-concentration half-cell is oxidised. The standard electrode potentials cancel. The concentration ratio is 100 and the electron count is two, giving the displayed potential of 0.0591 V. The cell reaction reduces the concentration difference.

Why not the others: (A) misses log ratio 100; (C) omits electron count two; (D) overlooks concentration difference.

Key point: Use concentration ratio and n=2.

Q29 / A Arrhenius temperature comparison

Answer: (A) 1.91.

Given: T_1=300 K, T_2=310 K, Ea=50000 J mol^{-1}.

Formula/Principle: $\ln(k_2/k_1)=Ea/R(1/T_1-1/T_2)$.

Step 1: Exponent ≈0.647.

Worked derivation:

$$\ln(k_{310}/k_{300}) = \frac{50000}{8.314}\left(\frac{1}{300} - \frac{1}{310}\right) \approx 0.647,$$

$$k_{310}/k_{300} = e^{0.647} \approx 1.91.$$

Why: The two-temperature Arrhenius ratio has $\ln(k_2/k_1)=(Ea/R)(1/T_1-1/T_2)\approx 0.647$, hence $\exp(0.647)\approx 1.91$. The factor is dimensionless and greater than one as temperature rises. Converting activation energy to joules is necessary because the stated gas constant uses joules.

Why not the others: (B) underestimates exponential sensitivity; (C) uses a smaller exponent; (D) exceeds the calculated factor.

Key point: Work with reciprocal absolute temperatures.

Q30 / D reaction quotient shift

Answer: (D) Forward until Q approaches K.

Why: Q<K means the present products-to-reactants activity ratio is below its equilibrium value. Forward reaction increases product relative to reactants until the equilibrium condition is approached.

Why not the others: (A) uses product identity instead of Q; (B) fixed T fixes K but not Q; (C) equilibrium is the condition Q=K, not Q<K.

Key point: Compare reaction quotient with equilibrium constant.

Mathematics

Q31 / C Row operations followed by an adjugate determinant

Answer: (C) 36.

Given: A is invertible of order three and $\det A = 3$.

Formula/Principle: Row addition preserves determinant; row scaling multiplies it by the scale factor; $\det(\operatorname{adj} B) = (\det B)^{n-1}$.

Step 1: After adding twice row 1 to row 2, the determinant remains 3.

Step 2: Scaling row 3 by −2 gives

$$\det B = -2(3) = -6 \neq 0.$$

Step 3:

$$\det(\operatorname{adj} B) = (-6)^2 = 36.$$

Worked derivation:

$$\det B \det(\operatorname{adj} B) = \det((\det B) I_3) = (\det B)^3 \Rightarrow \det(\operatorname{adj} B) = 36.$$

Why: The determinant sign changes at row scaling, but the adjugate identity then squares the result in three dimensions. These are distinct operations; taking the absolute value of the determinant would not in general replace the adjugate calculation.

Why not the others: (A) retains a negative sign after squaring; (B) stops at $\det B$; (D) takes only the magnitude of $\det B$.

Key point: Complete row operations before applying the matrix-order-dependent identity.

Q32 / D path count avoiding a point

Answer: (D) 8.

Given: Three R and three U steps; forbidden point (1,1).

Formula/Principle: Nallowed=C(6,3)−C(2,1)C(4,2).

Step 1: 20−2·6=8.

Worked derivation:

$$N = \binom{6}{3} - \binom{2}{1}\binom{4}{2} = 20 - 12 = 8.$$

Why: There are 6 choose 3 =20 paths overall. A path through (1,1) has 2 choose 1 ways to reach it and 4 choose 2 ways afterward, 12 forbidden paths. Subtract to get 8; giving the selected 8 valid paths.

Why not the others: (A) counts forbidden paths; (B) miscounts the complement; (C) understates valid paths.

Key point: Use product counting through the forbidden point.

Q33 / A composition domain

Answer: (A) $(-\infty,-\sqrt{2})\cup(\sqrt{2},\infty)$.

Given: $f(g(x))=\ln(x^2-2)$.

Formula/Principle: Logarithm argument $x^2-2>0$.

Step 1: $x<-\sqrt{2}$ or $x>\sqrt{2}$.

Worked derivation:

$$x^2-2>0 \iff x\in(-\infty,-\sqrt{2})\cup(\sqrt{2},\infty).$$

Why: Composition gives $\ln(x^2-2)$, whose argument must be strictly positive. Solving $x^2>2$ gives two open rays; equality is excluded because ln 0 is undefined.

Why not the others: (B) reverses the inequality and includes endpoints; (C) permits arguments below zero; (D) uses the domain of g rather than the composed logarithm.

Key point: Apply the outer logarithm restriction after composition.

Q34 / B modulus locus

Answer: (B) $x-y=2$.

Given: Points (1,1) and (3,−1).

Formula/Principle: $(x-1)^2+(y-1)^2=(x-3)^2+(y+1)^2$.

Step 1: $4x-4y-8=0$.

Worked derivation:

$$(x-1)^2+(y-1)^2=(x-3)^2+(y+1)^2 \iff x-y=2.$$

Why: For $z=x+iy$, equate squared distances to (1,1) and (3,−1). Simplification gives $-4x+4y+8=0$, hence $x-y=2$. The locus is the perpendicular bisector of the two points.

Why not the others: (A) has wrong orientation; (C) changes the constant; (D) is a shifted parallel line.

Key point: Equidistance becomes a linear equation after squares cancel.

Q35 / C symmetry of definite integral

Answer: (C) π.

Given: Integral from 0 to π.

Formula/Principle: $\int x\sin x\,dx=-x\cos x+\sin x$.

Step 1: Upper endpoint π; lower endpoint 0.

Worked derivation:

$$\int_0^{\pi} x\sin x\,dx=[-x\cos x+\sin x]_0^{\pi}=\pi.$$

Why: Integration by parts with $u=x$ and $dv=\sin x\,dx$ gives $-x\cos x+\sin x$. At π this is π, at zero it is zero.

Why not the others: (A) halves the boundary term; (B) doubles it; (D) overlooks the nonzero endpoint contribution.

Key point: Keep the boundary term in integration by parts.

Q36 / D foot of perpendicular

Answer: (D) (22/25,46/25).

Given: P=(1,2), normal=(3,4), signed offset=1.

Formula/Principle: H=P−[(3·1+4·2−10)/25](3,4).

Step 1: H=(22/25,46/25).

Worked derivation:

$$H = (1,2) - \frac{1}{25}(3,4) = \left(\frac{22}{25}, \frac{46}{25}\right).$$

Why: The signed line expression at P is 3+8−10=1. Subtract 1/(3²+4²) times the normal vector (3,4), yielding (22/25,46/25), the selected foot.

Why not the others: (A) is off the projection; (B) lies elsewhere on the line; (C) is not a normal displacement.

Key point: Project along the oblique line normal.

Q37 / A parameterised differentiability

Answer: (A) (2,−1).

Given: Piecewise join at x=1.

Formula/Principle: Continuity a+b=1; derivative match a=2.

Step 1: b=−1.

Worked derivation:

$$a + b = 1, \quad a = 2 \quad \Longrightarrow \quad (a, b) = (2, -1).$$

Why: Continuity first requires a+b=1. Equality of one-sided derivatives requires a=2, so b=−1. Both conditions are necessary; matching only value is insufficient.

Why not the others: (B) is continuous but slopes differ; (C) has matching derivative but a+b≠1; (D) is continuous but left slope −2.

Key point: Check value and slope at the joining point.

Q38 / B Mean deviation about the median

Answer: (B) 1.

Given: The ordered data are $1, 1, 4$, so the median is 1.

Formula/Principle: Average the absolute distances from the specified centre.

Step 1:

$$|1-1| + |1-1| + |4-1| = 3.$$

Step 2:

$$\mathrm{MD}_{\text{median}} = \frac{3}{3} = 1.$$

Worked derivation:

$$\mathrm{MD} = \frac{0+0+3}{3} = 1.$$

Why: The requested centre is the median, not the arithmetic mean of two. Changing the centre changes the distances and therefore the final mean deviation.

Why not the others: (A) gives mean deviation about the mean; (C) gives the arithmetic mean itself; (D) reports the sum rather than average of deviations.

Key point: Use the centre named in the question.

Q39 / C continuity parameter

Answer: (C) 2.

Given: Limit of [sin(ax)−2x]/x³.

Formula/Principle: sin(ax)=ax−a³x³/6+O(x⁵).

Step 1: Finite iff a−2=0.

Worked derivation:

$$\frac{\sin(ax)-2x}{x^3}=\frac{a-2}{x^2}-\frac{a^3}{6}+O(x^2);\quad a=2.$$

Why: Expand sin(ax)=ax−a³x³/6+higher terms. The coefficient of x must vanish for the quotient to avoid x⁻² divergence, so a=2. The resulting finite limit is −8/6.

Why not the others: (A), (B) and (D) leave a nonzero (a−2)x term, producing divergence upon division by x³.

Key point: Cancel lower-order terms before taking a higher-order limit.

Q40 / D Bayes with unequal priors

Answer: (D) 4/7.

Given: P(U₁)=1/3, P(R|U₁)=2/3; P(U₂)=2/3, P(R|U₂)=1/4.

Formula/Principle: Bayes P(U₁|R)=P(U₁)P(R|U₁)/P(R).

Step 1: P(R)=2/9+1/6=7/18.

Worked derivation:

$$P(U_1\mid R)=\frac{(1/3)(2/3)}{(1/3)(2/3)+(2/3)(1/4)}=\frac{4}{7}.$$

Why: The red joint probabilities are (1/3)(2/3)=2/9 and (2/3)(1/4)=1/6. Their sum is 7/18, so posterior U₁ is (2/9)/(7/18)=4/7. The observation favours U₁ despite its smaller prior because red is substantially more likely there.

Why not the others: (A) repeats the prior; (B) uses U₁ likelihood alone; (C) ignores unequal priors and urn compositions.

Key point: Normalise joint probabilities after the observation.

Q41 / A compound-angle identity

Answer: (A) 4/3.

Given: tan θ=1/2.

Formula/Principle: tan 2θ=2tanθ/(1−tan²θ).

Step 1: 1/(1−1/4)=4/3.

Worked derivation:

$$\tan 2\theta=\frac{2(1/2)}{1-(1/2)^2}=\frac{4}{3}.$$

Why: Apply tan 2θ=2t/(1−t²) with t=1/2: numerator 1, denominator 3/4, so 4/3. The acute condition fixes a consistent quadrant but the algebra already determines the value.

Why not the others: (B) inverts the result; (C) ignores the denominator; (D) doubles t incorrectly.

Key point: Use the full double-angle denominator.

Q42 / B Nonlinear homogeneous initial-value equation and positive branch

Answer: (B) $e\sqrt{3}$.

Given: $y' = (x^2 + y^2)/(xy)$, $x > 0, y > 0$, and y(1)=1.

Formula/Principle: Use $v = y/x$, so $y' = v + xv'$ and the right-hand side is $v + 1/v$.

Step 1:

$$v + xv' = v + 1/v \Rightarrow v\,dv = dx/x.$$

Step 2:

$$\frac{v^2}{2} = \ln x + C, \quad v^2 = 2\ln x + C_1.$$

Step 3: At x=1, v=1, so $C_1 = 1$. The positive branch is

$$y = x\sqrt{1 + 2\ln x}, \quad x > e^{-1/2}.$$

Step 4:

$$y(e) = e\sqrt{1+2} = e\sqrt{3}.$$

Worked derivation:

$$y = x\sqrt{1 + 2\ln x}, \qquad y(e) = e\sqrt{3}.$$

Why: The homogeneous substitution removes the repeated ratio y/x and reduces the equation to separable form. The given positivity chooses the square-root branch and also excludes the boundary where the original equation would have y=0 in its denominator.

Why not the others: (A) drops the integration constant fixed by y(1); (C) replaces the square root of three by two; (D) omits the square root.

Key point: Carry the initial value, sign branch and domain through a homogeneous substitution.

Q43 / C discriminant and parameter

Answer: (C) (1+√13)/2≤k<12/5.

Given: Polynomial has sum 2k, product k+3, vertex x=k.

Formula/Principle: Δ=4(k²−k−3)≥0; positive branch k≥(1+√13)/2.

Step 1: At x=3, p(3)=12−5k>0 and k<3.

Step 2: Hence lower endpoint included, upper excluded.

Worked derivation:

$$k \in \left[\frac{1+\sqrt{13}}{2}, \frac{12}{5}\right).$$

Why: Discriminant nonnegative and positive roots force k≥(1+√13)/2. To keep both roots below 3, the vertex k must lie below 3 and the polynomial at x=3 must be positive: 12−5k>0, so k<12/5. At the lower end the repeated root lies inside (0,3), while the upper end places a root at excluded x=3.

Why not the others: (A) includes arbitrarily large k; (B) wrongly excludes the valid repeated root and includes the endpoint 3; (D) makes the polynomial at 3 negative.

Key point: Combine discriminant, vertex position, endpoint signs and strict boundaries.

Q44 / D Minimum triangle area from positive intercepts

Answer: (D) 12.

Given: $a, b > 0$ and $2/a + 3/b = 1$.

Formula/Principle: Let $u = 2/a,\ v = 3/b$; then $u + v = 1$ and $uv \le 1/4$.

Step 1:

$$ab = \frac{6}{uv} \ge 24.$$

Step 2:

$$\text{Area} = \frac{ab}{2} \ge 12.$$

Step 3: Equality occurs for $u = v = 1/2$, giving $a = 4, b = 6$, whose line does pass through (2,3).

Worked derivation:

$$\min \frac{ab}{2} = \frac{1}{2}\frac{6}{1/4} = 12.$$

Why: Positive intercepts allow the AM–GM bound on two positive quantities with fixed sum. Checking equality shows the lower bound is attained by an actual line rather than only approached.

Why not the others: (A) confuses the area of a smaller rectangle with the triangle; (B) and (C) fall below the proved bound.

Key point: Convert intercept conditions to a fixed-sum product bound.

Q45 / B collinearity test

Answer: (B) 5.

Given: First two points define y=2x.

Formula/Principle: For y=10, x=10/2.

Step 1: k=5.

Worked derivation:

$$\frac{6-2}{3-1} = 2, \quad y - 2 = 2(x-1) \implies 10 = 2k \implies k = 5.$$

Why: The first two points have slope (6−2)/(3−1)=2. The line through (1,2) is y=2x; setting y=10 gives x=5. This is option B.

Why not the others: (A), (C) and (D) do not satisfy the line equation y=2x at y=10.

Key point: Use one line equation and substitute the third point.

Physics

Q46 / A limiting friction and impending slip

Answer: (A) 4 N opposite the applied force.

Given: Applied 4 N < limiting 6 N.

Formula/Principle: Horizontal equilibrium: f_s=Fapplied while at rest.

Step 1: f_s=4 N, opposite the force.

Worked derivation:

$$\sum F_x = 0 \implies f_s = F = 4\,\text{N} < 6\,\text{N}.$$

Why: For a stationary block static friction adjusts to match the applied horizontal force up to its limiting value. At 4 N, below the measured 6 N limit, force balance requires 4 N opposite the applied force.

Why not the others: (B) confuses the maximum with actual friction; (C) leaves unbalanced force; (D) would let the block accelerate before the observed threshold.

Key point: Static friction is self-adjusting below its limit.

Q47 / B position-dependent work

Answer: (B) −32/3 J.

Given: F=x²−4x, 0≤x≤4.

Formula/Principle: W=∫₀⁴F(x) dx.

Step 1: W=[x³/3−2x²]₀⁴=−32/3 J.

Worked derivation:

$$W = \int_0^4 (x^2 - 4x)\,dx = \left[\frac{x^3}{3} - 2x^2\right]_0^4 = -\frac{32}{3}\,\text{J}.$$

Why: The force is negative throughout the open interval 0<x<4, so work must be negative. Integrating gives [x³/3−2x²]₀⁴=64/3−32=−32/3 J. The zeros of force at the endpoints do not cancel the negative area accumulated between them.

Why not the others: (A) uses magnitude but loses sign; (C) assumes endpoint zeros imply zero area; (D) replaces the curved force profile by an incorrect constant.

Key point: The signed area under F(x) determines work.

Q48 / C escape speed ratio

Answer: (C) √3v/2.

Given: v²=2GM/R; target radius 4R.

Formula/Principle: ½mu²=GMm(1/R−1/4R).

Step 1: u²=(3/4)v².

Worked derivation:

$$u = \sqrt{\frac{2GM}{R}\left(1 - \frac{1}{4}\right)} = \frac{\sqrt{3}}{2}v.$$

Why: Energy conservation requires initial kinetic energy equal to the increase in gravitational potential from −GMm/R to −GMm/(4R). That difference is 3GMm/(4R). As v²=2GM/R, the required speed is √3v/2, below escape speed because the target is finite.

Why not the others: (A) is local escape speed at 4R, a different quantity; (B) assumes inverse distance scaling of launch speed; (D) omits the square root.

Key point: Separate local escape speed from finite-height launch energy.

Q49 / D Poisson-ratio response

Answer: (D) −0.0006.

Given: ν=0.30, longitudinal strain +0.002.

Formula/Principle: εtrans=−ν εlong.

Step 1: −0.30×0.002=−0.0006.

Worked derivation:

$$\varepsilon_\perp = -\nu\varepsilon_\parallel = -(0.30)(0.002) = -6 \times 10^{-4}.$$

Why: Poisson ratio ν=−εtrans/εlong. Substituting 0.30 and +0.002 gives εtrans=−0.0006: a tensile rod narrows sideways. Both strains are dimensionless; the negative transverse sign records contraction rather than an additional force.

Why not the others: (A) and (B) multiply by ten; (C) misses the opposite sign.

Key point: Poisson ratio contains a minus sign.

Q50 / A cooling comparison

Answer: (A) 3:2.

Given: Ambient 20°C, bodies 80°C and 60°C.

Formula/Principle: |dT/dt|=k(T−Ta).

Step 1: 60/40=3/2.

Worked derivation:

$$\frac{|\dot{T}_1|}{|\dot{T}_2|} = \frac{80-20}{60-20} = \frac{3}{2}.$$

Why: Newton cooling at the initial moment is proportional to the temperature excess over ambient: 80−20=60 K and 60−20=40 K. The rate ratio is 60:40=3:2.

Why not the others: (B) uses absolute temperatures incorrectly; (C) ratios the two object temperatures; (D) ignores different excesses.

Key point: Use temperature excess above the common ambient.

Q51 / B pressure from molecular impacts

Answer: (B) $Nm\langle c_x^2\rangle/L^3$.

Given: N molecules in cube volume L^3.

Formula/Principle: $P=Nm\langle c_x^2\rangle/V$.

Step 1: $V=L^3$.

Worked derivation:

$$P = \frac{N(mc_x^2/L)}{L^2} = \frac{Nm\langle c_x^2\rangle}{L^3}.$$

Why: One molecule transfers momentum $2mc_x$ at each collision, with collisions separated by $2L/|c_x|$, giving mean force mc_x^2/L. Sum N and divide by wall area L^2 to obtain $Nm\langle c_x^2\rangle/L^3$. A factor 1/3 is needed only when replacing component mean square by total speed mean square.

Why not the others: (A) misses a length; (C) double-counts the isotropy factor for a component already specified; (D) adds another unwarranted factor.

Key point: Derive pressure from momentum flux normal to the wall.

Q52 / C phase and energy at given displacement

Answer: (C) 3E/4.

Given: $E=\frac{1}{2}kA^2$ and $x=A/2$.

Formula/Principle: $U/E=x^2/A^2$, $K=E-U$.

Step 1: U=E/4, K=3E/4.

Worked derivation:

$$K = \frac{1}{2}k(A^2-x^2) = E\left(1-\frac{x^2}{A^2}\right) = \frac{3}{4}E.$$

Why: The spring potential at displacement x is $E(x^2/A^2)$. At A/2 it equals E/4; conservation leaves kinetic energy 3E/4. The speed is still nonzero away from the turning points, where kinetic energy would vanish entirely.

Why not the others: (A) gives potential energy; (B) assumes a linear energy split; (D) is kinetic energy only at equilibrium.

Key point: Energy varies as displacement squared.

Q53 / D dielectric with battery disconnected

Answer: (D) $V_0/3, U_0/3$.

Given: Q fixed; $C_{new}=3C_0$.

Formula/Principle: $V=Q/C$, $U=Q^2/(2C)$.

Step 1: $V_{new}=V_0/3$; $U_{new}=U_0/3$.

Worked derivation:

$$\frac{V_{\text{new}}}{V_0}=\frac{C_0}{3C_0}=\frac{1}{3}, \qquad \frac{U_{\text{new}}}{U_0}=\frac{C_0}{3C_0}=\frac{1}{3}.$$

Why: Disconnection fixes free charge Q. Insertion raises capacitance threefold, so $V=Q/C$ falls to one third and $U=Q^2/(2C)$ also falls to one third. The force drawing dielectric inward supplies the energy difference.

Why not the others: (A) reverses both changes; (B) incorrectly keeps voltage fixed after disconnection; (C) gives the battery-connected energy trend with inconsistent voltage.

Key point: Identify the fixed quantity before choosing energy formula.

Q54 / A force with vector field components

Answer: (A) $vB(\hat{y}-\hat{z})$.

Given: $v=v\,\hat{x}$; $B=B(\hat{y}+\hat{z})$; $q>0$.

Formula/Principle: $F=q(E+v\times B)=0$.

Step 1: $v\times B=vB(\hat{z}-\hat{y})$.

Worked derivation:

$$\mathbf{E}=-\mathbf{v}\times\mathbf{B}=vB(\hat{\mathbf{y}}-\hat{\mathbf{z}}).$$

Why: The magnetic part per unit charge is $v\times B=vB(\hat{z}-\hat{y})$. Electric and magnetic forces must cancel, so $E=-v\times B=vB(\hat{y}-\hat{z})$. This is a force-balance condition, not a magnetic-force-only direction question.

Why not the others: (B) adds to the magnetic force; (C) points partly parallel to B instead of cancelling $v\times B$; (D) leaves magnetic deflection.

Key point: Set $E+v\times B=0$ and resolve both components.

Q55 / B Induced surface charges and Gaussian flux

Answer: (B) $-q,\ -q,\ -q/\varepsilon_0$.

Given: Shell charge is $-2q$, cavity charge is $+q$; the external $-5q$ is excluded from the final Gaussian surface.

Formula/Principle: Field inside conducting material is zero; charge on the shell is conserved; net flux depends only on enclosed charge.

Step 1: A Gaussian surface within the metal around the cavity has zero flux, giving

$$q+Q_{\text{inner}}=0 \Rightarrow Q_{\text{inner}}=-q.$$

Step 2:

$$Q_{\text{outer}}=-2q-Q_{\text{inner}}=-q.$$

Step 3: The outer Gaussian surface encloses $+q-2q=-q$, hence

$$\Phi=-q/\varepsilon_0.$$

Worked derivation:

$$Q_{\text{inner}}=-q, \quad Q_{\text{outer}}=-q, \quad \Phi=-q/\varepsilon_0.$$

Why: The external charge can redistribute the outer-surface charge density but cannot change the total charge of this isolated shell. It also cannot contribute to the net flux of a surface that excludes it.

Why not the others: (A) assigns too much total charge to the shell; (C) leaves a nonzero enclosed charge inside a Gaussian surface in the metal; (D) reverses induced inner charge and includes the excluded external charge.

Key point: Apply metal equilibrium, charge conservation and Gauss's law to different surfaces.

Q56 / C resistor network node reduction

Answer: (C) 5 Ω.

Given: Two 6 Ω branches A–B; 2 Ω B–C.

Formula/Principle: RAB=(1/6+1/6)$^{-1}$; RAC=RAB+2.

Step 1: 3+2=5 Ω.

Worked derivation:

$$R_{AB} = \left(\frac{1}{6} + \frac{1}{6}\right)^{-1} = 3\,\Omega, \quad R_{AC} = 3 + 2 = 5\,\Omega.$$

Why: The equal parallel pair combines as (6×6)/(6+6)=3 Ω. Current then passes through the 2 Ω series resistor, giving 5 Ω overall. The common A–B endpoints establish parallel connection; the single B–C arm carries their combined current.

Why not the others: (A) omits series arm; (B) treats parallel pair as 2 Ω; (D) adds only one 6 Ω branch to 2 Ω.

Key point: Reduce nodes and series/parallel stages.

Q57 / D lens-mirror combination

Answer: (D) 50/3 cm left of lens.

Given: First lens object distance 30 cm, f=10 cm; mirror x=20 cm.

Formula/Principle: First image x=15; mirror image x=25; second pass lens object distance 25.

Step 1: Second pass 1/v=1/10−1/25=3/50.

Worked derivation:

$$v_1 = 15\,\text{cm}, \quad x_{\text{mirror image}} = 25\,\text{cm}, \quad v_2 = \left(\frac{1}{10} - \frac{1}{25}\right)^{-1} = \frac{50}{3}\,\text{cm left}.$$

Why: On the first pass the lens makes an image 15 cm to its right. The mirror lies 5 cm beyond it and forms a virtual image at x=25 cm behind the mirror. Returning rays approach the lens as if from an object 25 cm to its right; the lens then makes a real image 50/3 cm to its left.

Why not the others: (A) is mirror image distance from mirror, not final location; (B) is focal length; (C) stops at the first lens image.

Key point: Track the intermediate image through both optical elements and the return pass.

Q58 / A photocurrent saturation

Answer: (A) Current doubles; stopping potential unchanged.

Given: Frequency fixed, intensity doubled.

Formula/Principle: Isat∝photon flux; eVstop=hf−φ.

Step 1: Isat doubles, Vstop unchanged.

Worked derivation:

$$I_{\text{new}} = 2I_{\text{sat}}, \qquad eV_{\text{new}} = hf - \phi = eV_{\text{stop}}.$$

Why: Twice the photon flux ejects approximately twice as many photoelectrons per time, raising saturated current. Maximum electron kinetic energy depends on photon frequency, unchanged here, so stopping potential stays fixed.

Why not the others: (B) and (C) incorrectly tie stopping potential to intensity; (D) overlooks the larger electron count.

Key point: Intensity controls count; frequency controls maximum energy.

Q59 / C emission selection

Answer: (C) 4→3.

Given: $E_n = -13.6/n^2$ eV.

Formula/Principle: $\Delta E = 13.6(1/n_{low}^2 - 1/n_{high}^2)$, $\lambda = hc/\Delta E$.

Step 1: 4→3 has smallest ΔE.

Worked derivation:

$$\Delta E_{4\to3} = 13.6\left(\frac{1}{9} - \frac{1}{16}\right) < \Delta E_{3\to2}, \Delta E_{4\to2}, \Delta E_{2\to1}.$$

Why: Longest wavelength means smallest emitted energy. Hydrogen gaps shrink among high neighbouring levels: 4→3 has 13.6(1/9−1/16) eV, less than the other displayed drops.

Why not the others: (A) spans two levels and is larger; (B) has a larger adjacent gap lower in the ladder; (D) has the largest of these gaps.

Key point: Compare energy gaps, then invert for wavelength.

Q60 / D Charge from a change of mutual flux linkage

Answer: (D) Positive charge $3MI_0/(4R)$ in the defined circuit reference direction.

Given: $I_1(0) = I_0$ and $I_1(T/2) = I_0/4$; the secondary is purely resistive in the stated approximation.

Formula/Principle:

$$I_2 = \mathcal{E}_2/R = -\frac{M}{R}\frac{dI_1}{dt}, \quad q_2 = \int I_2\, dt.$$

Step 1:

$$\Delta I_1 = I_0/4 - I_0 = -3I_0/4.$$

Step 2: Integrate Faraday's relation over the specified interval:

$$q_2 = -\frac{M}{R}\Delta I_1 = \frac{3MI_0}{4R}.$$

Step 3: Since $dI_1/dt < 0$ throughout this interval, emf and secondary current are positive in the explicitly defined reference.

Worked derivation:

$$q_2 = -\frac{M}{R}[I_1(T/2) - I_1(0)] = \frac{3MI_0}{4R}.$$

Why: The total charge depends on the change of linked flux, not on a single instantaneous current or slope. Stating the signed linkage and circuit convention makes the direction unambiguous without requiring an unstated dot convention.

Why not the others: (A) reverses the Faraday sign; (B) uses the final primary current instead of its change; (C) assumes the quadratic current has fallen linearly by one half.

Key point: Integrate induced current using the signed flux-linkage change.
